

Logarithmic Lebesgue lower bounds for periodic nonuniform sampling

Partial-result packet for fa_banach_001

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Status. Substantial partial resolution of Conjecture 80 in Boche–Mönich [1]: the conjectured pointwise $c \log N$ lower bound holds for every periodic nonuniform perturbation of the integer sampling lattice. The general aperiodic complete-interpolating-sequence case remains open.

Source conjecture

Holger Boche and Ullrich J. Mönich, *Signal and System Approximation from General Measurements*, arXiv:1402.1092 (2014) [1]. In Conjecture 80, following Remark 26, they ask whether every ordered complete interpolating sequence $\{t_k\}_{k \in \mathbb{Z}}$ for $\mathcal{PW}_\pi^2$, with cardinal functions ϕ_k , satisfies

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \sum_{k=-N}^N e^{i\omega t_k} \widehat{\phi}_k(x) \right| dx \geq C \log N$$

for every $\omega \in [-\pi, \pi]$ and $N \in \mathbb{N}$. Their proved estimate has instead the maximum over all earlier truncation indices $1 \leq M \leq N$.

The cached TeX was faithfully compiled with a substitute article class because its publisher class was absent; the conjecture appears on page 18 of that local source PDF, where automatic numbering labels it Conjecture 5.3 rather than 80.

Theorem 5.1 shows that if the summation is restricted to a suitable subsequence of the natural numbers, we can have a convergent system approximation process if we use measurement functionals. Now the question arises if this is also true for pointwise sampling as analyzed in Section 3. Since in Theorem 3.7 we only have a $\limsup$ this could be the case. However, we have the following conjecture.

Conjecture 5.3. *Let $\{t_k\}_{k \in \mathbb{Z}} \subset \mathbb{R}$ be an ordered complete interpolating sequence for $\mathcal{PW}_\pi^2$. Then there exists a positive constant C_{12} such that*

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \sum_{k=-N}^N e^{i\omega t_k} \widehat{\phi}_k(\omega_1) \right| d\omega_1 \geq C_{12} \log(N) \quad (49)$$

for all $\omega \in [-\pi, \pi]$ and all $N \in \mathbb{N}$.

If this conjecture is true then the derivations in this work imply that a theorem such as Theorem 5.1 cannot hold for the sampling-based system approximation that was treated in Section 3. Because then, for every subsequence $\{M_N\}_{N \in \mathbb{N}}$ of the natural numbers and all ordered complete interpolating sequences $\{t_k\}_{k \in \mathbb{Z}} \subset \mathbb{R}$ we have

$$\limsup \left(\sup \left| (T_* f_*)(t) - \sum_{k=-M_N}^{M_N} f_*(t_k) (T_* \phi_k)(t) \right| \right) = \infty \quad (50)$$

Definitions and theorem

Fix an integer $q \geq 2$ and real numbers

$$0 = \alpha_0 < \alpha_1 < \cdots < \alpha_{q-1} < q.$$

For $m \in \mathbb{Z}$ and $0 \leq r < q$, define

$$t_{qm+r} = qm + \alpha_r. \quad (1)$$

These nodes are strictly increasing and $t_0 = 0$. Let $\phi_k \in \mathcal{PW}_\pi^2$ be their cardinal functions, normalized by

$$\phi_k(t_\ell) = \delta_{k\ell}, \quad \phi_k(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{\phi}_k(x) e^{ixt} dx.$$

Put

$$L_N(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \sum_{k=-N}^N e^{i\omega t_k} \hat{\phi}_k(x) \right| dx.$$

Theorem 1 (Periodic nonuniform case). *The sequence in (1) is a complete interpolating sequence for $\mathcal{PW}_\pi^2$. There is a constant $c = c(q, \alpha_1, \dots, \alpha_{q-1}) > 0$ such that*

$$L_N(\omega) \geq c \log(N+1)$$

for every $N \geq 1$ and every $\omega \in [-\pi, \pi]$.

Corollary 2 (Alternating nonuniform grids). *For every $0 < a < 2$, the nonuniform sequence*

$$t_{2m} = 2m, \quad t_{2m+1} = 2m + a$$

satisfies Conjecture 80. Except when $a = 1$, its consecutive gaps alternate between a and $2 - a$, so this is a genuinely non-equidistant family.

Proof intuition

Split the frequency interval into q equal cells. Under this splitting, the exponential basis associated with (1) becomes the ordinary Fourier basis on one cell, multiplied pointwise by a fixed Vandermonde matrix and harmless unimodular phases. Inverting that finite matrix gives the biorthogonal functions explicitly.

For a fixed evaluation frequency ω , restrict the Lebesgue integral to the same cell as ω . All residue classes modulo q have a large common block of truncation indices. Their contribution factors as a scalar Dirichlet kernel times a continuous coefficient whose value on the diagonal is exactly q . The non-common endpoints contribute only a uniformly bounded remainder. The local L^1 -norm of the Dirichlet kernel is logarithmic, and the bounded remainder cannot cancel that growth.

Proof

Partition $[-\pi, \pi)$ into cells

$$I_j = I_0 + s_j, \quad I_0 = [-\pi, -\pi + 2\pi/q), \quad s_j = \frac{2\pi j}{q}, \quad 0 \leq j < q.$$

Set

$$z_r = e^{2\pi i \alpha_r / q}, \quad V = (z_r^j)_{\substack{0 \leq j < q \\ 0 \leq r < q}}.$$

The z_r are distinct, so the Vandermonde matrix V is invertible.

The exponential system is a Riesz basis

Use the cell-splitting map

$$(\mathcal{J}f)(y) = (f(y + s_0), \dots, f(y + s_{q-1})), \quad y \in I_0.$$

For $k = qm + r$,

$$\mathcal{J}(e^{it_k \cdot})(y) = e^{iqmy} e^{i\alpha_r y} v_r, \quad v_r = (1, z_r, \dots, z_r^{q-1})^\top.$$

Thus the split system is obtained from the orthogonal vector Fourier basis $\{e^{iqmy} e_r : m \in \mathbb{Z}, 0 \leq r < q\}$ by multiplication with

$$W(y) = V \operatorname{diag}(e^{i\alpha_0 y}, \dots, e^{i\alpha_{q-1} y}).$$

Both $W(y)$ and $W(y)^{-1}$ are uniformly bounded on I_0 . Consequently $\{e^{it_k x}\}_{k \in \mathbb{Z}}$ is a Riesz basis of $L^2[-\pi, \pi]$. By the complete-interpolation/Riesz-basis equivalence recalled in the source [1, Theorem 11], the nodes form a complete interpolating sequence for $\mathcal{PW}_\pi^2$.

Explicit biorthogonal functions

Let

$$C = q(V^\top)^{-1} = (c_{jr})_{0 \leq j, r < q}. \quad (2)$$

We claim that the Fourier transforms of the cardinal functions are characterized cellwise by

$$\widehat{\phi}_{qm+r}(y + s_j) = e^{-i(qm+\alpha_r)y} c_{jr}, \quad y \in I_0. \quad (3)$$

Indeed, for $t_{qn+s} = qn + \alpha_s$, cell splitting gives

$$\begin{aligned} & \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{it_{qn+s}x} \widehat{\phi}_{qm+r}(x) dx \\ &= \frac{1}{2\pi} \int_{I_0} e^{iq(n-m)y} e^{i(\alpha_s - \alpha_r)y} \sum_{j=0}^{q-1} z_s^j c_{jr} dy. \end{aligned}$$

By (2), the sum is $q\delta_{sr}$. When $s = r$, orthogonality of $\{e^{iqmy}\}$ on the interval of length $2\pi/q$ then makes the last display δ_{nm} ; when $s \neq r$, it is zero before integration. Hence (3) is the unique biorthogonal family.

Dirichlet core of every truncation

Fix $\omega \in I_j$, write $\omega = y_\omega + s_j$, and restrict the integration variable to $x = y + s_j \in I_j$. Put $u = \omega - x$ and

$$b_{jr} = z_r^j c_{jr}, \quad A_j(u) = \sum_{r=0}^{q-1} b_{jr} e^{i\alpha_r u}.$$

From (3),

$$K_N(\omega, x) := \sum_{k=-N}^N e^{i\omega t_k} \widehat{\phi}_k(x) = \sum_{r=0}^{q-1} b_{jr} e^{i\alpha_r u} \sum_{\substack{m \in \mathbb{Z} \\ |qm+r| \leq N}} e^{iqmu}. \quad (4)$$

Since $C^\top = qV^{-1}$,

$$A_j(0) = \sum_{r=0}^{q-1} V_{jr} C_{jr} = (VC^\top)_{jj} = q.$$

There are finitely many j , so some $\eta > 0$, chosen also with $\eta \leq \pi/q$, satisfies

$$|A_j(u)| \geq q/2 \quad (0 \leq j < q, |u| \leq \eta). \quad (5)$$

Write $N = q\ell + s$, $0 \leq s < q$. The integer ranges in the inner sums of (4) all contain the same consecutive block

$$J_N = \begin{cases} \{-\ell, \dots, \ell-1\}, & s \leq q-2, \\ \{-\ell, \dots, \ell\}, & s = q-1. \end{cases}$$

Each residue class has at most two further endpoint indices. Therefore

$$K_N(\omega, x) = A_j(u)P_N(u) + R_{N,j}(u), \quad P_N(u) = \sum_{m \in J_N} e^{iqmu}, \quad (6)$$

where

$$|R_{N,j}(u)| \leq B := 2 \max_j \sum_{r=0}^{q-1} |b_{jr}| \quad (7)$$

for every N, j, u .

Local logarithmic lower bound

If a consecutive integer set has cardinality d , then

$$\left| \sum_{m=a}^{a+d-1} e^{iqmu} \right| = \frac{|\sin(dqu/2)|}{|\sin(qu/2)|}.$$

The elementary Dirichlet-kernel estimate

$$\int_0^\eta \frac{|\sin(dqu/2)|}{|\sin(qu/2)|} du \geq c_0 \log(d+1) - C_0 \quad (8)$$

holds with constants depending only on q, η . For completeness, $|\sin(qu/2)| \leq q|u|/2$, and after the change of variables $v = dqu/2$, the lower bound follows by summing $\int_{m\pi+\pi/6}^{m\pi+5\pi/6} |\sin v|v^{-1} dv$ over $m \leq d\eta q/(2\pi)$.

For every ω , one of the two one-sided intervals of length η starting at ω lies in the same cell I_j ; at a cell boundary, choose either adjacent cell. On that interval, (5)–(7) and the triangle inequality give, for all sufficiently large N ,

$$\begin{aligned} 2\pi L_N(\omega) &\geq \int_0^\eta (|A_j(u)||P_N(u)| - |R_{N,j}(u)|) du \\ &\geq \frac{q}{2} (c_0 \log(|J_N| + 1) - C_0) - B\eta \\ &\geq c_1 \log(N+1) - C_1, \end{aligned}$$

uniformly in ω . This is at least $c_2 \log(N+1)$ once N is large.

For each of the finitely many remaining N , the function $\omega \mapsto L_N(\omega)$ is continuous and strictly positive: a finite nontrivial linear combination of the biorthogonal functions cannot vanish in L^2 . Shrinking c_2 covers those N as well and proves Theorem 1. Corollary 2 is the case $q = 2, \alpha_0 = 0, \alpha_1 = a$. $\square$

Verification notes

- The identity $VC^T = qI$ is the key noncancellation check: it forces the coefficient of the Dirichlet core to equal q on the diagonal, independently of the cell.
- Every residue-class truncation differs from the common block J_N by at most two indices. This is why the remainder in (6) is bounded independently of N .
- The optional checker in the packet’s code directory verifies the Vandermonde/dual identities and numerically evaluates sample Lebesgue functions for alternating and three-periodic grids. No numerical output is used in the proof.
- The proof gives constants depending on the fixed periodic pattern. It does not assert a uniform constant over degenerating patterns where two α_r ’s approach one another modulo q .

Upgrade attempts, novelty, and limitations

The source’s first two named conjectures are proved later in the same paper and were rejected as extraction noise. Conjecture 80 is genuine. Direct attempts to upgrade the source’s Szarek lemma fail because that lemma is intrinsically a maximal-partial-sum statement. General lower bounds for projection norms control a supremum over ω , whereas Conjecture 80 requires every ω and every truncation. Finite or summable perturbations of a periodic pattern were also considered, but the dual Riesz basis changes nonlocally, so the uniformly bounded endpoint remainder used here no longer follows.

A bounded search on 11 August 2026 used the paper title, Conjecture 80, complete interpolating sequences, periodic nonuniform sampling, Lebesgue constants, and logarithmic lower bounds. It found the authors’ later *Strong Divergence for System Approximations* [2], which strengthens equidistant and adaptive-kernel results but does not establish the arbitrary nonuniform kernel estimate above. It also found literature on periodic nonuniform reconstruction and on complete interpolating sequences, but no exact pointwise L^1 lower bound matching this theorem. Novelty confidence is therefore moderate, not definitive.

The unresolved step is genuinely aperiodic: for a general complete interpolating sequence, splitting into finitely many cells does not produce a fixed finite matrix, and no common Dirichlet core with bounded remainder is available. The full Conjecture 80 remains open in this packet.

Human-review recommendation

Validity confidence is high. A reviewer should check the transpose convention in (2), the normalization q in the biorthogonality calculation, and the common-block indexing for $N = q\ell + s$. Those are the only places where a phase or endpoint error could affect the theorem. Specialist literature review is recommended before a public novelty claim.

References

- [1] H. Boche and U. J. Mönich, *Signal and System Approximation from General Measurements*, [arXiv:1402.1092](#) (2014).
- [2] H. Boche and U. J. Mönich, *Strong Divergence for System Approximations*, [arXiv:1505.03057](#) (2015).